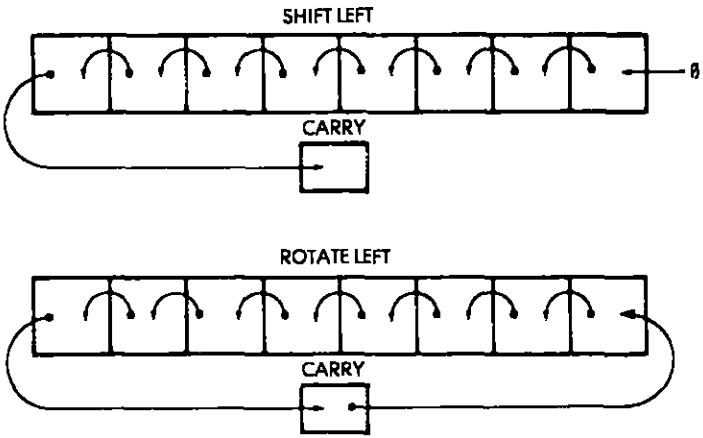
The *ALU* performs arithmetic and logic operations. A special register equips one of the inputs of the ALU, the left input here. It is called the accumulator. (Several accumulators may be provided.) The accumulator may be referenced both as input and output (source and destination) within the same instruction.

The ALU must also provide *shift* and *rotate* facilities.

A shift operation consists of moving the contents of a byte by one or more positions to the left or to the right. This is illustrated in Figure 2.3. Each bit has been moved to the left by one position. The details of shifts and rotations will be presented in the next chapter.



Note: Some Shift and Rotate instructions do not include the Carry.

Fig. 2.3: Shift and Rotate

The shifter may be on the ALU output, as illustrated in Figure 2.2, or may be on the accumulator input.

To the left of the ALU, the *flags* or *status register* appear. Their role is to store exceptional conditions within the microprocessor. The contents of the flags register may be tested by specialized instructions, or may be read on the internal data bus. A *conditional* instruction will cause the execution of a new program, depending on the value of one of these bits.

The role of the status bits in the Z80 will be examined later in this chapter.